

Non-Functional Requirements

- **NFR1 – Security:** Passwords are hashed (PBKDF2), sessions are managed via cookies with a 7-day expiration and sliding expiration, and HTTPS is enforced.
- **NFR2 – Usability:** A unified layout (shared EduCoreLayout) provides an easy-to-use interface for both teachers and students.
- **NFR3 – Maintainability:** The codebase follows the MVC pattern with EF Core Migrations, so any model change translates into a clear, trackable migration.
- **NFR4 – Data Integrity:** Foreign keys and relationship constraints (Restrict on certain relationships to avoid multiple cascade paths) ensure data consistency.
- **NFR5 – Scalability:** The architecture supports adding new roles (like Assistant) or new features (online payment) without a major schema overhaul, since supporting tables (Card, Assistant, TeacherAssistant) already exist in preparation.
- **NFR6 – Performance:** EF Core navigation properties are used to reduce the number of queries needed when fetching related data.
- **NFR7 – Reliability:** Data validation is enforced at the model level (StringLength, Required, Range) before persisting to the database.